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June 16, 2022 By [Hanna Welch](https://filewhopper.com/author/anna/) (<https://filewhopper.com/author/anna/>)

Ways to Use External Hard Drive as RAM



There are two types of computer memory, namely physical and virtual memory. The RAM chips attached to your motherboard represent the physical memory, whereas the virtual memory comes from the reserved space within your hard drive (<https://filewhopper.com/blog/ssd-or-hdd-which-one-should-you-use/>). When a program is not in use but open, the Windows system uses the virtual memory to store its data.

Now, if you have an old computer with little RAM, less than 4GB, to be precise, then you need to upgrade your device by increasing virtual memory using your external hard drive. There are a lot of untrustworthy theories about boosting this and that on your computer to make it better. In most cases, the misleading of readers comes from the wrong presentation of information, whether intentional or unintentional. For instance, in this case, increasing virtual memory will not increase the memory inside the RAM chips.

Actually, the RAM chips' capacity cannot be changed unless they are taken back to the manufacturing company for reconstruction. Therefore, this article is focusing on adding virtual memory using an external hard drive. We will also show you how to free up RAM. That way, you can get the best results after installing external RAM for laptop. So, how does using an external hard drive increase RAM and improve performance?

Well, with more virtual memory, a computer is able to multitask, therefore completing tasks faster. If a program is open but idle, the computer can shift its data to virtual memory, thus relieving the physical RAM. Thus, when properly configured, your PC will improve its performance. Now, for you to understand things better, let's first explain how virtual memory works before diving deep into the setup procedure.

Virtual memory is kept in a system file called pagefile or paging file. Virtual memory is not accessible to the user and can only be accessed under two circumstances: when the physical RAM is overloaded and when there are programs that are open but idle. So, the first scenario is common among old PCs that

have little physical RAM. When your computer is running low on physical memory, the newly generated data from programs that are being opened still needs to go somewhere.

You may wonder, why not just increase physical RAM? Remember, old computers have a limited RAM volume, i.e. on some old PCs, motherboards can only take up to 2GB of RAM. Therefore, you are only left with virtual memory. Anyway, back to our case: so, once the physical RAM is left with no space, Windows turns to paging files, which opened programs will use as if they were physical memory. When an open program is minimized or not in use for some time, its data is moved from RAM to virtual memory, hence space for other applications and important system operations. This, in turn, helps the whole computer system to run efficiently.

Virtual memory plays a pivotal role in keeping your PC operations smooth. Although some websites discourage the use of paging files, this can prove to be a bad idea in the long run. If there is no paging file in your system, then you will likely experience slowdowns, errors, freezes, and even crashes leading to data loss. Even if you have a large amount of physical memory, leaving breathing space for it by implementing virtual memory will help keep your computer healthy. Note that, if your paging file is small, you might also experience these problems, hence the need to know the right size to set your virtual memory because setting it too high may also lead to issues.

There are other facts that you need to consider when it comes to virtual memory before setting it up. One of the major misinterpreted concepts about virtual memory is the thought that it operates the same way as physical RAM. However, virtual memory can never substitute RAM, and it is slower. Even if used on the fastest solid-state drive, it will remain slower than physical memory. Therefore, if your computer is slow due to not having enough physical RAM memory, having a well-setup paging file will only help in stabilizing and bettering its performance but will definitely not make it much faster.

There isn't a specific amount of virtual memory that can be labelled as best for any PC. Therefore, the amount of virtual memory you need depends on your user needs. To be precise, the amount of virtual memory you need depends on the amount of physical RAM you have and how much memory you usually use. For instance, if you have 6GB of RAM installed and your max memory usage is capped at 9GB, it would be best to set your paging file memory to 4GB. This means, in total, you will have 10GB, giving you an extra GB to be on the safe side.

How to Use a Hard Drive as RAM on Windows

Here is how you can set up your external hard drive as RAM to enjoy a more stable computer:

Step 1

1. Access the **Start menu** and right-click on **Computer**.
2. Select **Properties** and then choose **Advanced system settings**.
3. If prompted, enter the admin password.

Step 2

1. Under System Properties, select the **Advanced** tab.
2. Choose the **Settings** button under the Performance category.

Step 3

1. Select the Advanced tab and choose the **Change button** located near the bottom of the window.
2. Find and uncheck the box labelled **Automatically manage paging file size for all drives**.

Step 4

1. Find and choose the external hard drive from the list.
2. Select **System managed size**, and your PC will allocate virtual memory on the external hard drive.
3. To allocate a customized virtual memory volume, select **Custom** and enter the minimum and maximum amount of your memory and choose Set.
4. Click on **OK** and then restart your computer for the settings to take effect.

FileWhopper - Send Large Files Fast and Securely

Sending large files and folders online can be stressful. Most of the trusted services require paying heavy fees before accessing their services. However, with FileWhopper, you get to access premium features with no subscription needed: it's a pay-as-you-go service, so you pay only when you need to transfer large files or folders, and the price is based on their size.

You can literally **send data of any size securely** to up to 8 recipients and choose it to be stored for up to 90 days. The process is simple to follow, and there are safety measures employed to protect your data.

Here is how it works:

- Visit the [FileWhopper \(https://filewhopper.com/?utm_source=fw_blog&utm_medium=inarticle_link&utm_campaign=ways-to-use-external-hard-drive-as-ram\)](https://filewhopper.com/?utm_source=fw_blog&utm_medium=inarticle_link&utm_campaign=ways-to-use-external-hard-drive-as-ram) website, select the file or folder you want to share and upload it to FileWhopper to get a price quote that is based on the size of the uploaded data.
- Upload the file or folder you wish to send using the FileWhopper app.
- Once the upload has started, it's the right time to share the download link and the password for decrypting your content with your recipient(s) as the app supports simultaneous upload and download.
- Upon closing the upload app, it automatically gets deleted from your computer. The download link can also be found in your personal account. And please double-check that you have saved your password.

If you are looking to save money when performing regular file transfers, we have some good news for you. The business version of FileWhopper is in the pipeline, and it will surely become a game changer thanks to its innovative transfer package technology.

With [FileWhopper Business \(https://filewhopper.com/business/?utm_source=fw_blog&utm_medium=inarticle_link_business&utm_campaign=ways-to-use-external-hard-drive-as-ram\)](https://filewhopper.com/business/?utm_source=fw_blog&utm_medium=inarticle_link_business&utm_campaign=ways-to-use-external-hard-drive-as-ram), you prepay for a 100GB, 500GB or 5TB package and use that amount of data to perform as many transfers as you want whenever you like – there are no time limits or other subscription restrictions, and you can even share your account with co-workers. This solution lowers the cost of transferring your files and folders when compared to using a personal FileWhopper account, and its reusable app makes the transfer process easier by allowing multiple simultaneous uploads.

FileWhopper Business can save your time, money and energy when sharing formidable amounts of data, which is why lots of users are excited about the upcoming release.

How to Free Up RAM: 3 Useful Tips

Although adding RAM external storage in Windows will help boost performance, freeing up your PC's RAM will ensure more satisfying results. If your computer is [running low on memory](https://filewhopper.com/blog/how-to-speed-up-your-pc-with-more-ram-a-quick-tutorial/), there are a few things you can do. This section will explain the different ways to free up RAM on a Windows machine.

1. Check RAM Usage with Windows Tools

The first thing you should do is check for the programs that use the most RAM on your computer. Fortunately, you can easily do so on a Windows PC.

Here is how:

1. Press the **Win + Shift + Esc** combination to bring up the Task Manager.
2. When it opens, go to the **Memory section** on the right side of the window.
3. Here, you should see all the running apps and the amount of RAM they use.
4. Note the apps that use the most RAM, as those might be the culprits affecting your machine's performance.
5. You can also check out the **Performance tab** at the top of the window if you want more details.

2. Uninstall or Disable Unneeded Software

Since you now have a list of the apps that consume the most RAM, check if there are any useless ones that you can get rid of. If there are, then follow these steps to remove them from your machine:

1. Press the **Windows logo key + I** combination to open the Settings app.
2. Go to the **Apps section**.
3. Scroll down the **Apps & features** section to find the app you want to remove.
4. Select it and then click on the **Uninstall button**.
5. Follow any other required instructions to uninstall the program.
6. Repeat these steps for the other useless apps, and reboot your computer.
7. Your machine's performance should now improve due to the freed-up RAM.

3. Update Your Apps

Some programs usually suffer from memory leaks which can easily deplete your machine's RAM storage. Luckily, updating the apps can quickly fix the issue. Therefore, check if there are any updates for your installed apps and install all of them.

[fwa-calculator]

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Gramhum

Again, depending on what computer you buy, you may not be able to upgrade your RAM. So consider this carefully the next time you buy a new computer, and make sure it's either upgradeable or comes equipped with as much RAM as you think you'll need.

brent

can any size external HD be used
any better than the rest or all the same
i can get a 1 tb from scan disk for \$45

Aj macchia

I am trying to play minecraft and some other video games on an ho laptop and it has 4gb ram and the article says u can use an extended storage for ram is how much storage does that take away from 500gb of extended storage?

Aj macchia

Hp labtop I meant to say sorry

Computer RAM

This is such a nice blog. I recently made my own PC and the best thing is that you can easily make it when you put your heart to it.

FileWhopper

Thanks for your feedback! We're glad you liked our blog.

Maria Torphy